VICTORY 2027

# Natural Vegetation

LECTURE-2 Geography

Sameer Sir



# Topics To Be Covered

1. Types of Vegetation
   - Deciduous
   - Desert
   - Littoral
2. Tropical Deciduous
3. Tropical Desert Forests
4. Littoral Forests



# Natural Vegetation

- *Importance of forests.*
- *Types of vegetation; distribution and correlation with their environment (tropical evergreen, tropical deciduous, tropical desert, littoral and mountain).*
- *Forest conservation – afforestation, reforestation, social forestry, agroforestry and Van Mahotsav*

# Telegram

https://t.me/icsebysameersir



In a Tropical Rain Forest, trees reach heights of up to 60 metres, forming a continuous upper canopy, while the forest floor remains dark and sparse. What is the primary ecological driver for this stratified (layered) structure?

A. Intense competition among plants for soil nutrients

B. Adaptation to resist heavy surface runoff and soil erosion

C. Intense competition to capture maximum sunlight

D. Maximizing water absorption during rainless periods

Despite having high species diversity and valuable hardwoods like Mahogany and Rosewood, Tropical Rain Forests in India are not commercially exploited on a large scale. Which combination of factors best explains this?

1. Trees do not grow in pure stands (heterogeneous distribution).
2. Wood is light and soft, unsuitable for timber.
3. Dense undergrowth and marshy ground hinder transport.
4. High timber extraction costs due to lack of demand.

A. 1 and 2 only

B. 1 and 3 only

C. 2 and 4 only

D. 1, 3, and 4 only

grows

2. Tropical Evergreen forests thrive in regions receiving an annual rainfall of:

(a) Less than 50 cm

(b) 50 cm to 100 cm

(c) 100 cm to 200 cm

(d) More than 200 cm

5. Which forest type forms a dense canopy that prevents sunlight from reaching the ground?

(a) Thorn and Scrub Forests

(b) Tropical Evergreen Forests

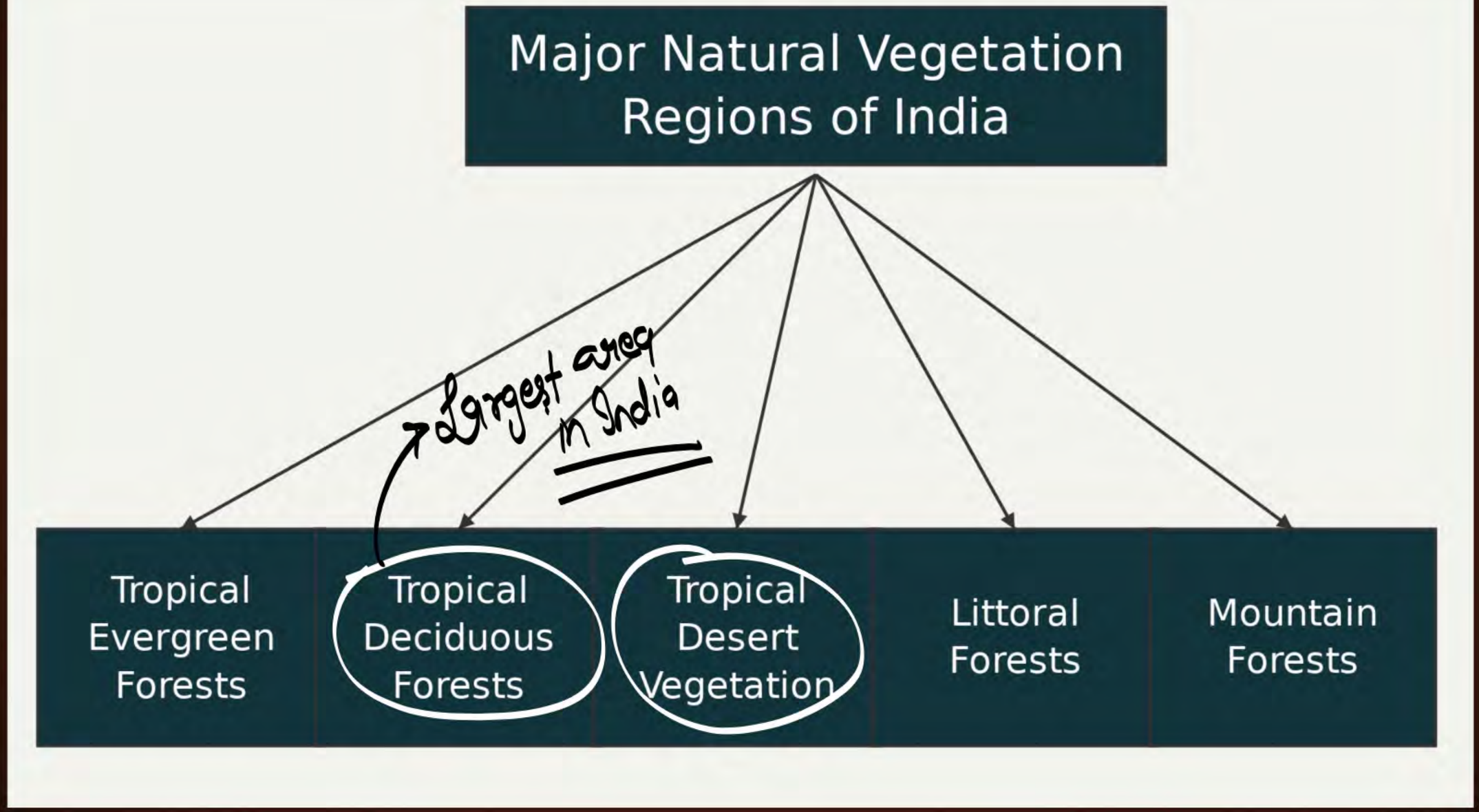
(c) Mountain Forests

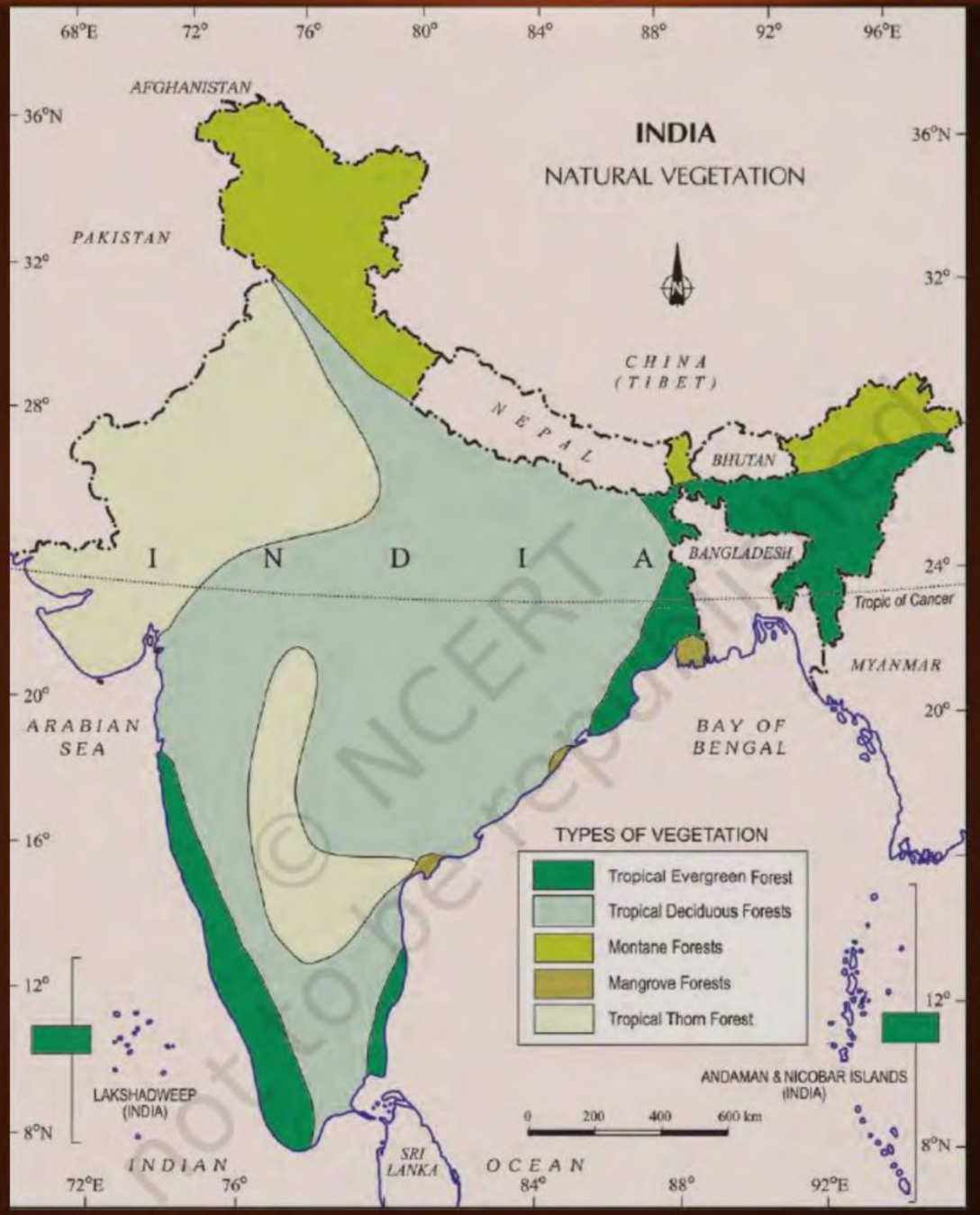
(d) Deciduous Forests








Major Natural Vegetation Regions of India
Largest area in India
Tropical Evergreen Forests
Tropical Deciduous Forests
Tropical Desert Vegetation
Littoral Forests
Mountain Forests


INDIA
NATURAL VEGETATION
AFGHANISTAN
PAKISTAN
CHINA (TIBET)
NEPAL
BHUTAN
BANGLADESH
MYANMAR
I N D I A
Tropic of Cancer
ARABIAN SEA
BAY OF BENGAL
INDIAN OCEAN
SRI LANKA
LAKSHADWEEP (INDIA)
ANDAMAN & NICOBAR ISLANDS (INDIA)
TYPES OF VEGETATION
Tropical Evergreen Forest
Tropical Deciduous Forests
Montane Forests
Mangrove Forests
Tropical Thorn Forest
0 200 400 600 km
68°E
72°
76°
80°
84°
88°
92°
96°E
36°N
32°
28°
24°
20°
16°
12°
8°N



# 2. Tropical Deciduous Forests

- These forests are also known as the monsoon forests.
- They are the most widespread forests in India.
- Based on the availability of water, these forests are further categorised into two types:
  - the moist deciduous forests;
  - the dry deciduous forests.

# 2. Tropical Deciduous Forests

## (i) The Moist Deciduous Forests

**(a) Climatic Conditions:** Such forests are found in areas with moderate annual rainfall of 100 cm to 200 cm and the mean annual temperature of 24° C and 27° C and humidity percentage of 50 to 80.

**(b) Distribution:** These forests occur in the north-eastern part of the Peninsula, along the foothills of the Himalayas and eastern slopes of the Western Ghats. They occupy a sizeable area in Uttar Pradesh, Maharashtra, Karnataka and Tamil Nadu and have an area bigger than the Evergreen forests.

# 2. Tropical Deciduous Forests

## (c) Characteristic Features

(i) The trees in these forests shed their leaves from six to eight weeks during spring and early summer when the shortage of water is acute. Further, the sub-soil water is not enough for the trees to keep their leaves all the year around.

(ii) A particular species can be found over a large area.

# 2. Tropical Deciduous Forests

(iii) They are commercially most exploited forests of India. The common trees of such forests are sal, teak, arjun, shisham, mahua, mulberry, palas, semul and sandalwood.

↳ चंदन


SAL
(Shorea rebusta)
TEAK
(Tectona grandis)
ARJUN
(Terminalia arjuna)
SHISHAM
(Dalbergia sissoo)
MAHUA
(Madhuca longifolia)
MULBERRY
(Morus alba)
PALAS
(Butea monosperma)
SEMUL
(Bombax ceiba)
SANDALWOOD
(Santalum album)


Sal
Teak
Arjun
Shisham
Mahua
Mulberry
Palas
Semul
Sandalwood

# 2. Tropical Deciduous Forests

(d) **Economical Value:** Tropical Deciduous Forests are commercially the most exploited. Besides providing valuable timber, they provide various other products. Sandalwood found in these forests is in great demand in India and abroad.

(e) **Environmental Correlation:** These forests shed leaves during dry seasons, helping conserve water and soil nutrients.



# 2. Tropical Deciduous Forests

## (ii) Dry Deciduous Forests

**(a) Climatic Conditions:**

- a mean annual temperature of 23°C to 27°C,
- annual rainfall between 70 cm to 100 cm and
- humidity between 51 to 58 per cent.

**(b) Distribution:** These forests are found in the rainier parts of the Peninsular Plateau and the plains of Bihar and Uttar Pradesh.

# 2. Tropical Deciduous Forests

**(c) Characteristic Features:** These forests thrive between moist deciduous forests (in the east) and tropical thorn forests (in the west).

- In the northern Indian plains and in the areas of higher rainfall in the Peninsular Plateau, these forests have open stretches in which teak and other trees are interspersed with patches of grass.
- During the dry season, the trees in these forests shed their leaves completely and give the forests a look of a vast grassland with naked trees.
- The common trees of these forests are teak, tendu, sal, rosewood, amaltas, bel, khair, axlewood, etc.

1




**TEAK**
*(Tectona grandis)*

2




**TENDU**
*(Diospyros melanoxylon)*

3




**SAL**
*(Shorea robusta)*

4




**ROSEWOOD**
*(Dalbergia latifolia)*

5




**AMALTAS**
*(Cassia fistula)*

6




**BEL**
*(Aegle marmelos)*

7




**KHAIR**
*(Acacia catechu)*

8




**AXLEWOOD**
*(Anogeissus latifolia)*

# 2. Tropical Deciduous Forests

(d) **Economical Value:** The trees of these forests provide timber, fruits and other useful products. Large tracts of these forests have been cleared for agricultural activities.

(e) **Environmental Correlation:** Adapted to dry conditions, these forests prevent soil erosion and support local wildlife.




Saturated

# 3. Tropical Desert Forests

These are also known as Tropical Thorn Forests.

(a) **Climatic Conditions:** rainfall less than 50 cm ,

- the mean annual temperature ranges between 25°C to 27°C and
- has humidity below 47 per cent.



# 3. Tropical Desert Forests

(b) **Distribution:** These forests are chiefly distributed in south-western Punjab, Haryana, Uttar Pradesh, central and eastern Rajasthan, Madhya Pradesh, Chhattisgarh, Gujarat, parts of Maharashtra, Karnataka and Andhra Pradesh.

# 3. Tropical Desert Forests

## (c) Characteristic Features:

Scarcity

कंटीले पौधे

Thorns

(i) These forests have Xerophytic vegetation. Due to paucity of rainfall, the trees are stunted with large patches of coarse grasses.

↳ Small

(ii) These forests have trees which have adapted themselves to survive in drought like conditions and are called xerophytes.

For example, the acacia or babool trees have developed long tap roots that can reach deep, ground water resources and therefore, can survive drought like conditions.





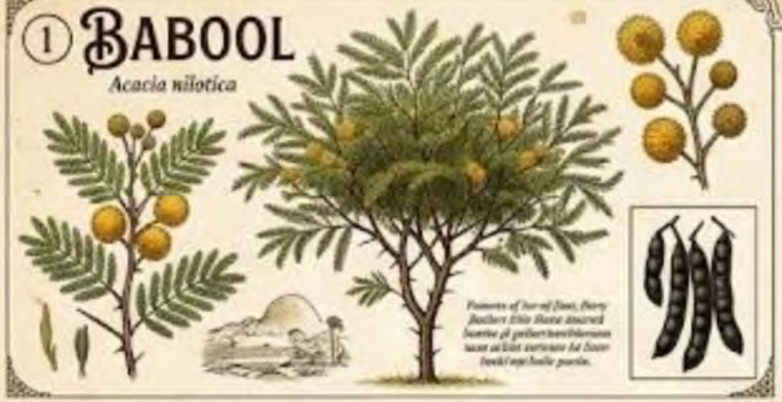
# 3. Tropical Desert Forests

(iii) In these forests, plants remain leafless for most part of the year and look like scrub vegetation.

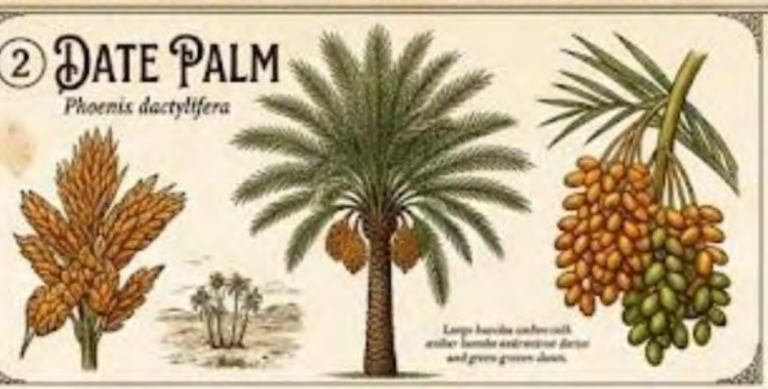
The important trees found here include babool or acacia, date palm, ber, khair, neem, khejri, kanju, cactii, kokko, etc.

# REPRESENTATIVE TREES & PLANTS OF ARID REGIONS

## ① BABOOL
*Acacia nilotica*



## ② DATE PALM
*Phoenix dactylifera*



## ③ BER
*Ziziphus mauritiana*



## ④ KHAIR
*Senegalia catechu*

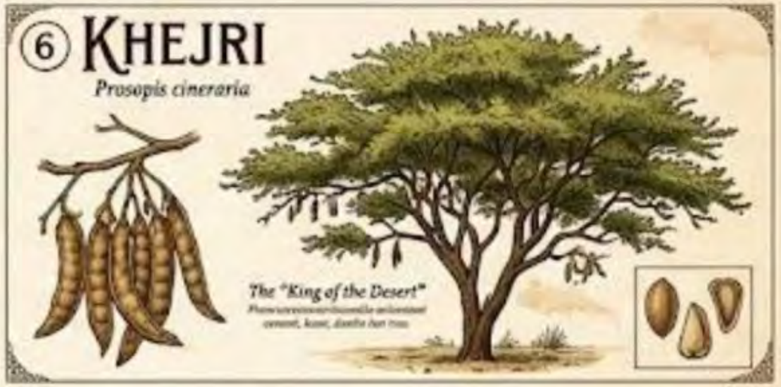


## ⑤ NEEM
*Azadirachta indica*



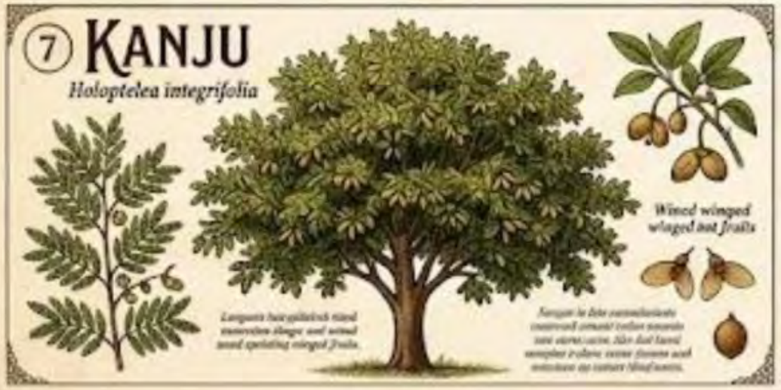
Neem seeds used

## ⑥ KHEJRI
*Prosopis cineraria*



The "King of the Desert"

## ⑦ KANJU
*Holoptelea integrifolia*



Winged winged nut fruits

## ⑧ CACTII
*Cactaceae spp.*



## ⑨ KOKKO
*Albizia lebbeck*



Rattling seeds, seed pods

# 3. Tropical Desert Forests

(d) **Economical Value:** Berfruit is eaten raw or made into pickle or beverages. Its timber is hard, strong, tough and durable. It is used to make legs for bedsteads, boat ribs, agricultural implements, charcoal, etc.

- Babool bark and gum have medicinal value.
- Date Palm is eaten raw and also used as an astringent, as a decoction, syrup or paste for sore throat, cold, fever, etc.
- Neem bark and roots have medicinal properties. Neem oil, leaves and extracts are used to manufacture health and beauty products. It is also used as an insecticide.

(e) **Environmental Correlation:** These plants are adapted to conserve water and survive extreme conditions, preventing desertification.

# 4. Littoral or Tidal Forests → Mangrove Forests



**Exposed Stilt Roots During Low Tide**

- Littoral or Tidal forests are one of the most specialised tropical natural vegetation types.
- India has a variety of wetland habitats.
- Wetlands are lands between terrestrial and aquatic system where the water table is usually at or near the surface of land.

**(a) Climatic Conditions:** These forests thrive in areas where temperature remains between 26° C to 29° C and the rainfall is not more than 200 cm .



# 4. Littoral or Tidal Forests

(b) **Distribution:** These forests are found in wet marshy areas, in river deltas, in tidal or other swampy areas and along the sea coasts.

- They are chiefly distributed in the deltas of large rivers on the eastern coast and in pockets on the western coast in saline swamps of Sunderbans in West Bengal and coastal areas of Andhra Pradesh and Odisha.

# 4. Littoral or Tidal Forests

Stilt → Stand

## (c) Characteristic Features:



(i) They have mainly evergreen species of varying density and height, usually associated with wetness.

(ii) The tree trunks are supported by a number of stilt like roots which remain submerged under water during high tide and can be seen during low tide.

(iii) They have profuse growth with tangle of climbers, which is an adaptation for survival in soft and shifting mud.

# 4. Littoral or Tidal Forests

(iv) These forests have breathing roots called pneumatophores.

- Because of waterlogged conditions, the roots are deprived of oxygen during high tides.
- So, some mangrove roots extend vertically above the ground.
- These vertical roots have pores which enable the trees to breathe when other roots remain sub merged under water during high tide.



# 4. Littoral or Tidal Forests

- The important trees include keora, amur, bhara, rhizophora, screw pines, canes and palms, sundari, agar, etc.
- They are found in the Andaman and Nicobar Islands and West Bengal.
- Other areas of significance include the Mahanadi, the Godavari and the Krishna deltas.
- These forests have Sundari trees, after the name of which these forests are known as 'Sunderbans' in the Ganga Delta.

# PLANTS OF THE SUNDARBANS MANGROVE AND ASSOCIATED FORESTS

## KEORA (*Sonneratia apetala*)



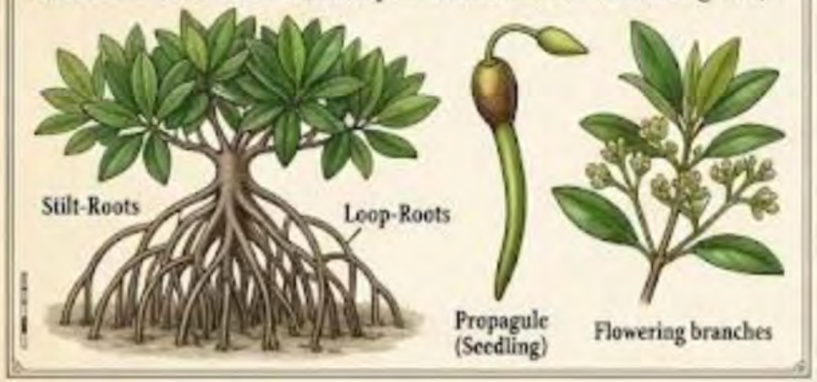

## AMUR (*Acer ginnala* - included species variation)



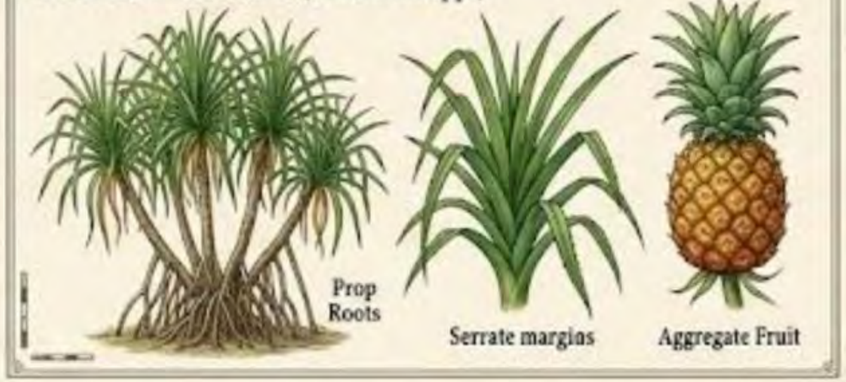

## BHARA (*Terminalia bellirica*)




## RHIZOPHORA (*Rhizophora mucronata - true mangrove*)



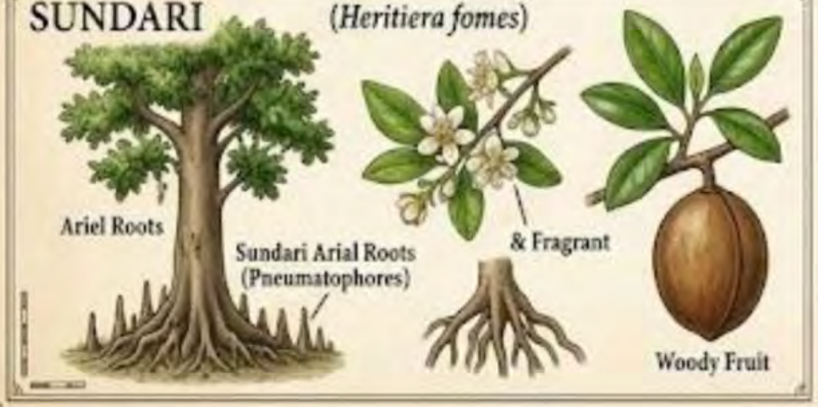

## SCREW PINES (*Pandanus spp.*)




## CANES AND PALMS (*Calamus spp. & Phoenix spp.*)




## SUNDARI (*Heritiera fomes*)




## AGAR (*Aquilaria malaccensis - Agarwood tree*)




## ETC. (*Other Mangrove Associates*)





# 4. Littoral or Tidal Forests

(d) **Economical Value:** Mangrove trees are utilised for fuel whereas sundari trees provide hard durable timber.

(e) **Environmental Correlation:** Mangroves protect coastlines from erosion and support marine life.





6. Trees with stilt roots, breathing roots (pneumatophores), [redacted] are characteristic of:

(a) Tropical Evergreen Forests

(b) Tidal (Mangrove) Forests

(c) Desert Vegetation

(d) Sub-tropical Montane Forests

8. Desert and Semi-Desert vegetation is mostly found in regions with rainfall less than:

(a) 25 cm

(b) 50 cm

(c) 100 cm

(d) 150 cm

9. Trees in Thorn and Scrub forests have long, deep taproots mainly to:

(a) Anchor the tree against strong winds

(b) Reach deep underground water tables

(c) Absorb nutrients from lower rock layers

(d) Support heavy foliage

12. Teak and Sal are the two most economically valuable species of which vegetation type?

(a) Tropical Deciduous Forests

(b) Mountain Forests

(c) Mangrove Forests

(d) Thorn and Scrub Forests

2025 PYQ's

H.W.

Q.1 State any three differences between tropical deciduous & tropical desert Vegetation? → (3)

Q.2 Give any two reasons for the great diversity of Vegetation found in India? → (2)

Q.3 How do forests help in the maintenance of our quality? → (1)

(2023) PYQ

Q.1 State any two ways by which forest protect the environment?

Q.2 Mention two ways in which tropical Desert vegetation adapted to it's environment?

# Thank You